

Lecture 13 (2026-08-02)

Quick Summary

The lecture covers decision-making regarding special orders and make-or-buy (outsourcing) decisions, emphasizing the distinction between relevant and irrelevant costs, the impact of capacity, and the qualitative risks of outsourcing. It also covers vertical integration, the decision to drop a product or service, and resource utilization under constraints. Additionally, the lecture covers the Theory of Constraints, "Process Further" decisions, and Cost-Volume-Profit (CVP) analysis, including Margin of Safety and Target Sales calculations.

Key Definitions

- **Special Order:** An order for a specific quantity or price, often outside the normal scope of business (e.g., a one-time discounted offer).
- **Upstream Costs:** Costs related to the production of a product (e.g., R&D, Product Development).
- **Downstream Costs:** Costs related to marketing, distribution, and customer service after the product is made.
- **Relevant Costs:** Costs that change based on a specific decision. In special orders, these are usually only the incremental (variable) costs.
- **Downside Risk:** A situation where the potential negative consequences of an action (e.g., a safety failure or brand damage from outsourcing) outweigh the potential benefits.
Vertical Integration: The extent to which a company owns the supply chain.
 - **Full Integration:** Complete control over the production process, selling, and marketing.
 - **Trade-off:** Higher control vs. potential for suppliers to provide higher quality items at lower costs.
- **Resource Utilization:** An analysis of how best to use a resource that is available in limited supply (e.g., shelf space in a grocery store or machine hours in a factory). These are typically short-term decisions.
- **Theory of Constraints:** A management tool for dealing with constraints. It identifies "bottlenecks" in the production process (factors such as machine time, labor, or delivery) that limit "throughput."
- **Sunk Cost:** A cost that has already been incurred and cannot be recovered; it is irrelevant to future decision-making.
- **Margin of Safety:** The portion of sales that contributes to the profit of the company; essentially, how much sales exceed the break-even point. It has no upper limit.

Worked Examples

Special Order Analysis (Sunset Airlines) * Scenario: * Total Capacity: 900 seats * Normal Load: 700 passengers * Available Capacity: 200 seats * Special Order Request: 150 seats * Special Order Price: \$125 * Normal Ticket Price: \$275 * Total Cost per passenger: \$137 (includes fixed costs) * Incremental/Variable Cost (food/drinks): \$6.50 *
Step-by-Step Calculation: 1. **Determine Capacity:** Check if there is enough excess capacity. (200 available > 150 needed). 2. **Identify Relevant Costs:** Ignore the total cost of \$137 because it includes costs already incurred. Focus only on the incremental cost of \$6.50. 3. **Calculate Incremental Profit:** \$125 (Offer) - \$6.50 (Incremental Cost) = \$118.50 (Approx. \$119) per passenger. * **Decision Logic:** Since there is excess capacity and the offer price exceeds the incremental cost, the order should be accepted.

Outsourcing Analysis (Airplane Maintenance) * Scenario: Determining whether to outsource maintenance to a third party. * **Analysis:** * **Benefit:** Potential cost reduction. * **Risk:** "Downside Risk" – if quality fails, it leads to safety issues and brand damage. * **Internal Impact:** Outsourcing may damage employee morale, lead to rumors of further cuts, or result in labor strikes.

Make or Buy Analysis (Birdie Maker Putters) * Scenario: Birdie Maker must decide whether to manufacture putters internally or purchase them from a supplier. * **Data:** * Offer Price (to buy): \$25 * Manufacturing Costs (Variable): \$17

(calculated as 17, 5, 9 components) * Fixed Manufacturing Overhead: \$9.50 (incurred regardless of decision) * Purchase Price (Base): \$15.50 * **Calculation Methods:** * **Method 1 (Include Fixed Cost):** * Cost to Make: \$17 (Variable) + \$9.50 (Fixed) = \$26.50 * Cost to Buy: \$15.50 (Purchase) + \$9.50 (Fixed) = \$25.00 (Wait, note: in the specific example provided, the calculation resulted in 26.5 vs 29 for the "buy" option comparison). * **Method 2 (Exclude Fixed Cost):** * Compare \$17 (Make) vs. \$15.50 (Buy). * **Decision Logic:** Both methods yield the same result. Because the \$9.50 fixed overhead is incurred regardless of the decision, it is an **irrelevant cost**. It can be added to both sides or removed from both sides without changing the final decision. * **Final Decision:** If cost to make (\$26.5) < cost to buy (\$29), the company should make.

Drop Product Analysis (Mud and Snow Tires) * Scenario: Decide whether to stop selling the "Mud and Snow" tire line. * **Data:** * **With Mud and Snow:** Sales = \$175,500; Fixed Overhead = \$28,000 * **Without Mud and Snow:** Sales = \$150,000; Fixed Overhead = \$26,000 * **Step-by-Step Calculation:** 1. **Calculate Contribution Margin:** * With product: \$175,500 (Sales) - (Variable Costs) = \$6,500 Contribution. * Without product: \$150,000 (Sales) - (Variable Costs) = \$4,500 Contribution. 2. **Identify the Gap:** By dropping the product, sales drop by \$25,500, but fixed costs only decrease by \$2,000. 3. **Determine Impact:** The contribution to the company's profit would drop from \$6,500 to \$4,500. * **Decision Logic:** Do not drop based solely on the "loss" shown on the individual product's statement. The product is currently covering \$4,500 of fixed costs that would otherwise remain even if the product were discontinued. Also, consider qualitative factors like "complete range" of products (e.g., customers won't buy bedsheets if pillows aren't offered).

Resource Utilization (Pro vs. Tour Models) * Scenario: BirdieMaker has a constraint on machine time. * **Data:** * **Pro Model:** Higher profit, less machine time required. * **Tour Model:** Lower profit, more machine time required. * **Calculation:** * **Pro Model:** \$300 contribution per machine hour. * **Tour Model:** \$200 contribution per machine hour. * **Decision Logic:** Since the Pro model generates more contribution per unit of the *limited resource* (machine time), the company should prioritize the Pro model.

Process Further Analysis (Furniture) * Scenario: Determine whether to sell furniture "as is" (unassembled) or process it further (assembled). * **Data:** * **Unassembled:** Sale Price = \$150; Production Cost = \$100 * **Assembled:** Sale Price = \$225; Production Cost = \$145 * **Step-by-Step Calculation:** 1. **Identify Incremental Revenue:** \$225 (Assembled) - \$150 (Unassembled) = \$75. 2. **Identify Incremental Cost:** \$145 (Assembled) - \$100 (Unassembled) = \$45. 3. **Calculate Net Gain:** \$75 - \$45 = \$30. * **Decision Logic:** Since the incremental revenue (\$75) is greater than the incremental cost (\$45), the product should be processed further. The original cost of production (\$100) is a **sunk cost** and does not impact the decision.

Cost-Volume-Profit (CVP) Calculation * Scenario: Calculate contribution and PV ratio for a company producing 1,000 units. * **Data:** * Units: 1,000 * Selling Price: \$80 * Direct Material Cost: \$20 * Other Variable Cost: \$30 * Fixed Costs: \$10,000 * **Step-by-Step Calculation:** 1. **Calculate Variable Cost per Unit:** \$20 (Material) + \$30 (Other) = \$50. 2. **Calculate Contribution per Unit:** \$80 (Price) - \$50 (Variable Cost) = \$30. 3. **Calculate Total Contribution:** \$30 $\times$ 1,000 = \$30,000. 4. **Calculate Total Sales:** \$80 $\times$ 1,000 = \$80,000. 5. **Calculate PV Ratio:** (\$30,000 / \$80,000) $\times$ 100 = 37.5%.

Margin of Safety (MOS) Calculation * Scenario: Determine the margin of safety for a company. * **Data:** * Selling Price: \$100 * Variable Cost: \$40 * Total Units Sold: 700 * Break-even Point: 300 units * **Step-by-Step Calculation:** 1. **Calculate Contribution per Unit:** \$100 (Price) - \$40 (Variable Cost) = \$60. 2. **Determine Break-even Point:** Total Fixed Cost / Contribution per Unit (e.g., \$18,000 / \$60 = 300). 3. **Determine Excess Units:** 700 (Total) - 300 (Break-even) = 400 units. 4. **Calculate Margin of Safety (Value):** 400 (Excess Units) $\times$ \$100 (Price) = \$40,000. * **Decision Logic:** The Margin of Safety identifies how far the current sales are from the loss-making point (the break-even point).

Cost in Financial Statements (Screenshot) | | | --- | --- | Revenues | | | Less: Cost of Goods Sold | | | Gross Margin | | | Less: Admn and S & D Expenses | | | Operating Profit | |

Formulas & Rules

Special Order Decision Rule: Only accept if:

1. There is excess capacity.
2. The offer price exceeds the *incremental* (variable) costs.

- **Opportunity Cost Calculation:** If capacity is **not** available, the cost of the special order must include the "lost" profit from the regular customer who is being displaced (e.g., if a seat is taken from a \$275 customer for a \$125 special order, the "cost" includes the \$150 difference).
- **Make or Buy Calculation Rule:** Fixed costs that are incurred regardless of the decision are **irrelevant**. You can add them to both options or exclude them from both; the decision remains identical.
- **Resource Utilization Rule:** When a resource is limited, prioritize products based on **Contribution Margin per unit of the limited resource** (e.g., CM per machine hour) rather than total profit per unit.
- **Process Further Rule:** Proceed with further processing only if **Incremental Revenue > Incremental Cost**.
- **Profit Calculation:** Profit = Sales - Cost (where Cost = Fixed Cost + Variable Cost).
- **Contribution Formula:** Contribution = Sales - Variable Cost.
- **Contribution/Profit Link:** Contribution = Profit + Fixed Cost.
- **Profit Volume (PV) Ratio:** (Contribution / Sales) $\times 100$.
- **Relationship Rule:** Variable Cost % + Profit Volume Ratio % = 100%.
- **Break-even Point (Units):** Fixed Cost / Contribution per Unit.
- **Break-even Point (Sales):** Fixed Cost / Profit Volume (PV) Ratio.
- **Break-even Point (Currency):** (Fixed Cost / Contribution per Unit) $\times$ Selling Price per Unit.
- **Margin of Safety (Calculation 1 - Value):** $\text{Margin of Safety} = \frac{\text{Profit}}{\text{Profit Volume (PV) Ratio}}$
- **Margin of Safety (Calculation 2 - Units):** $\text{Margin of Safety} = \frac{\text{Profit}}{\text{Contribution per Unit}}$
- **Sales to earn a desired profit (Value):** $\frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{Profit Volume (PV) Ratio}}$
- **Sales to earn a desired profit (Quantity/Units):** $\frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{Contribution per Unit}}$

Tricky Logic & Traps

- **The "Total Cost" Trap:** In special order questions, students often try to compare the offer price to the total cost (including fixed costs). You must only compare the offer to the **incremental/variable cost** unless the production of that specific order requires new, additional fixed costs.
- **Fixed Cost Irrelevance:** In "Make vs. Buy" decisions, the fixed cost (like a \$9.50 overhead) is often included in both scenarios. Whether you include it in your math or not, it does not change the choice. The decision is driven by the difference in variable costs.
- **The "Individual Loss" Trap:** When deciding to drop a product, a "loss" on a specific product's line item does not mean it's a loss for the company. The product may still be contributing significantly to covering common fixed costs. You must look at the **impact on total profit**.
- **The "Unit Profit" Trap:** In resource utilization, do not use "profit per unit." If machine time is limited, you must use **Contribution Margin per unit of limited resource**. A product that is more profitable per unit might be a worse choice if it takes significantly more time to produce.
- **Quality and Branding:** Not all decisions are pure math. If a "Make or Buy" decision involves a "safety-critical" function or a core brand promise (like a "complete range" of products), the "Downside Risk" may outweigh the cost savings.
- **Sunk Cost in Processing:** In "Process Further" decisions, the original manufacturing cost is a **sunk cost**. Students are often tripped up by large initial costs; however, only the *change* in cost vs. the *change* in revenue determines the decision.
- **PV Ratio Integrity:** To ensure calculation accuracy, the **Variable Cost % plus the Profit Volume Ratio % must equal 100%**. If it does not, a mistake was made in calculating either the variable cost or the contribution.
- **The "Negative COGS" Trap:** When calculating turnover ratios (like Gross Profit), ensure that **Cost of Goods Sold (COGS)** is entered as a **positive number** in the denominator. Even if the calculation for Gross Profit involves subtracting COGS from sales, the value of COGS itself is positive.
- **Margin of Safety "Limit":** Note that the **Margin of Safety has no upper limit**. It is simply the difference between actual sales and the break-even point.

Exam Pointers

- **Identify the Variable:** When calculating profit for a special order, ignore "fixed costs" that are already being incurred. Only include costs that "change" because of the order.
- **Capacity vs. Opportunity:** If the question states that capacity is full, you must subtract the "lost" contribution of the regular customer from the special order's profit.
- **Downside Risk:** If a question asks about the risk of outsourcing a "safety-critical" task, the answer is almost always that the risk is too high to outsource.
- **Resource Selection:** When multiple products compete for limited resources (time, space, labor), always calculate the **Contribution per Unit of Limited Resource**.
- **Processing Further:** If a question asks about "processing further," ignore the original production cost (it's a sunk cost). Focus only on whether the extra revenue outweighs the extra cost of the next step.
- **CVP Check:** Always check that your Variable Cost % and your Profit Volume (PV) Ratio add up to 100% to ensure your units and pricing logic are aligned.
- **Margin of Safety Check:** Use the difference between Total Sales and Break-even Sales to find the Margin of Safety. It can be expressed as a dollar amount or a number of units.
- **Target Sales Choice:** Use the **Profit Volume (PV) Ratio** to find the target sales in terms of **value** (currency), and **Contribution per Unit** to find the target sales in terms of **quantity** (units).

Sources

- Make or Buy and Special Order.docx
- FMA-ratios -2026 jan to june (1).pptx
- [shot_00-45-00.png] (Theory of Constraints Slide)
- [shot_01-05-35.png] (Cost in Financial Statements)